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Scalable enzymatic polymerization and low-temperature post-polymerization of poly(butylene succinate): Process parameters and application

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ABSTRACT

Poly(butylene succinate) is a semicrystalline bio-based and biodegradable polymer with good processability. However, using metal-based transesterification catalysts during chemical synthesis can be a drawback, especially for the biomedical and food packaging sector. An alternative approach was herein employed, using immobilized *Candida antarctica* Lipase B as a biocatalyst in a solvent-free process to synthesize PBS in line with green chemistry. The key process parameters (reaction temperature, pressure, time) were studied regarding the pre-polymer's molecular weight, thermal properties, and morphology. Thanks to the process's simplicity, scaling up followed, and a quantity of 20 g of PBS ($\overline{M_n}$ 1700 g·mol^{-1}) was received, thus filling the relevant gap in the open literature. The scaled-up grade was subsequently submitted to different sustainable post-polymerization methods (SSP, melt post-polymerization) to increase the molecular weight while maintaining the green character of the final product. The post-polymerized PBS ($\overline{M_n}$ 3000 g·mol^{-1}) was used as a carrier for the naturally occurring antioxidant flavonoid naringin. The release profile of naringin from the PBS nanocarrier was assessed, proving the enzymatically synthesized and upgraded PBS an effective matrix for high-purity encapsulation systems for the first time.

1. Introduction

Poly(butylene succinate) or PBS is a semicrystalline, thermoplastic polyester presenting glass transition temperature of −30 °C, melting point of 114 °C, thermal stability up to 220 °C and decomposition temperature above 390 °C. Thanks to its thermal properties and structure, PBS presents increased flexibility and satisfactory processability, being thus appropriate for film applications [1], e.g., in biomedicine (e.g. in soft tissue repair [2–4]) and the food packaging sector [5–8]. Especially for the latter, according to the European Commission, packaging waste increased by >20% over the last ten years and is forecast to soar by another 19% by 2023 [9]. PBS emerges as a promising alternative for polyolefins (mainly PP), considering its green character (bio-based and biodegradable) and its approval as a food contact material [10]. PBS has recently been introduced in the active food packaging sector too. PBS films modified with quercetin, a natural polyphenolic antioxidant, have been tested as antimicrobial and antioxidant food packaging materials [11], while PBS films filled with kesum, a new antimicrobial active agent, have been submitted to *in vivo* direct food contact analysis in a chicken fillet package [12]. In these studies, the incorporation of active agents and film formation was achieved *via* solution casting method. PBS has also been submitted to melt extrusion and compression moulding at relatively low temperatures and short times in a work aiming at preparing bioactive packaging films [13]. More precisely, melt extrusion was conducted in a twin screw extruder under mild conditions (135 °C, for 3 min, 40 rpm) to disperse the active agent CD-lim inclusion complex in a PBS matrix (melt mixing). CD-lim inclusion complex consists of limonene, a volatile essential oil with antimicrobial properties, encapsulated in β-cyclodextrins (β-CD) to protect limonene from oxidative degradation. The composite was placed in a compression moulding press at 130 °C, thus forming packaging films.

PBS conventional production requires metal-based transesterification catalysts (e.g., titanium (IV) *tert*-butoxide, titanium (IV)

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Table 1
Literature data on PBS enzymatic synthesis.

Ref.	Enzyme	Solvent	T (°C)	Range of achieved MW(g·mol^{-1})	
				$\overline{M_w}$	$\overline{M_n}$
[19]	N435	–	85	1094	851
[6]	N435	Isooctane	50	2000	–
[25]	N435	–	60	2550	1700
[24]	N435	–	90	4000	1400
[26]	N435	Diphenyl ether	80	–	840–2550
[27]	N435	Diphenyl ether	75	–	3910
[28]	N435	Diphenyl ether	80	11,520	4463–6017
[29]	N435	Diphenyl ether	60–90	–	2000–7000
[23]	N435	–	130	23,600	11,700
[29]	N435	–	80–95	38,000	27,340
[22]	N435	–	95	44,000	–
[30]	N435	Toluene	95	73,000	–
[20]	N435	Toluene	100–120	130,000	81,250

isobutoxide, titanium (IV) isopropoxide, tin (II) 2-ethylhexanoate, antimony (III) oxide, zirconium (IV) *n*-butoxide, antinomy (III) *n*-butoxide, hafnium (IV) *n*-butoxide, bismuth (III) neo- decanoate, germanium (IV) oxide), among which titanium-based ones are mainly used [14,15]. The most common industrial PBS synthesis routes include the use of the aforementioned catalysts in metal amounts between 100 and 1500 ppm, e.g., Ti 100–360 ppm, Sb 350–450 ppm, and Sn 450–1500 ppm [14]. However, these metal contents can be a significant drawback, especially for the food packaging industry, considering that according to the compostable and biodegradable polymers packaging standards, heavy metals should not exceed the maximum allowable level of 150 mg kg^{-1} (i.e., 150 ppm) on dry substance [16]. An alternative approach to producing high molecular PBS is the combination of polycondensation with a finishing step of chain extension [17]. A chain extender with two functional groups reacts with the terminal —OH or —COOH of PBS and couples two PBS chains. However, the biosafety and biodegradability of PBS can be negatively affected and thus, impair the use of the chain-extended PBS as a food-contacting material [8].

The most common methods to remove metal-based catalyst residues from polymers, especially when destined for food packaging applications, include solvent extraction and leaching. However, these methods require several steps and the use of many organic solvents and acids. Modern approaches that have been found effective, such as using a high-pressure CO_2/water binary system as a green solvent to dissolve and remove zinc-based catalysts from polymers, present high operation costs and need to be further optimized [16].

An alternative approach, totally avoiding chemical catalysts, is enzymatic polymerization. In enzymatic polymerization, non-toxic enzymes are used as catalysts, and they do not have to be completely removed from the final product, as they are harmless for humans and the environment. Additionally, any undesirable yellowing (mainly caused by titanium-based catalysts) and side reactions (e.g., the dehydration of 1,4 butanediol towards THF) during PBS conventional synthesis, are avoided since enzymatic polymerization is conducted at low temperatures and enzymes present high enantio-, chemo- and regioselectivity [18]. To be noted that enzymatically synthesized PBS can be considered truly green, combining green properties (e.g., biological origin and biodegradability) and a sustainable production process.

Enzymatically synthesized PBS grades with a wide range of MW, from oligomers ($\overline{M_n}$ 800 g·mol^{-1}) [19] to high molecular weight grades ($\overline{M_n}$ 81,000 g·mol^{-1}) [20], have been reported in the open literature (Table 1). The by-product removal, strongly affecting the reaction equilibrium, has been indicated as one of the most crucial parameters of the process. In this context, high-boiling point diphenyl ether is the most commonly used solvent, permitting the application of a high vacuum to remove polycondensation by-products. However, using various solvents (e.g., diphenyl ether for solution polymerization, chloroform for filtration, methanol for precipitation) impedes the enzymatic polymerization processes' scaling up [21]. Additionally, the requirement of high vacuum (i.e., 10 mmHg) [22] and temperatures (>90 °C) [20,23] for prolonged reaction times leads to increased MW but increases the operating costs and energy consumption, thus opposing the green chemistry principles. An alternative approach, permitting the process scale-up, has been suggested by the authors' group and included using low-boiling point solvents (e.g., isooctane), which are easy to remove from the system. At the same time, mild conditions were applied (40–60 °C, atmospheric pressure), but the achieved molecular weights and melting points were low ($\overline{M_w}$ 2000 g·mol^{-1} and T_m 78 °C) [6]. In a recent paper, PBS enzymatic polymerization was conducted in a solventless system, favoring scaling-up, at a "pre-melting" temperature (i.e., 90 °C), but the achieved molecular weights were found to be relatively low ($\overline{M_n}$ up to 1500 g·mol^{-1}, $\overline{M_w}$ up to 4000 g·mol^{-1}) [24]. The main drawbacks of this process, probably related to the low molecular weight values, are the co-existence of melted and solid oligomers' chains impeding the uniform mixing of the reacting mass and the possible monomers' loss before esterification at 90 °C considering also the applied high vacuum (0.1 atm).

Therefore, it seems crucial to conduct enzymatic polymerization at mild conditions and combine it with appropriate finishing steps to increase molecular weight and improve thermal properties, such as melting temperature. The authors' group has already applied solid-state post-polymerization (SSP) or bulk/melt post-polymerization in different enzymatically synthesized polyesters (e.g., PE 8.12, PE 8.14, PBS), and the results have been more than encouraging. Indicatively, Kanelli et al. [27] synthesized enzymatically in diphenyl ether PBS and submitted it to SSP under vacuum at temperatures close to the prepolymer melting point (T_m-T_{SSP} = 5–10 °C). The process was held in two stages, pre-crystalllization and SSP, and resulted in a $\Delta\overline{M_n}$ of 67%, with the $\overline{M_n}$ reaching the value of 6500 g·mol^{-1} (initial $\overline{M_n}$. 3910 g·mol^{-1}, Table 1). Gkountela et al. [6] synthesized enzymatically in the low boiling point isooctane, PBS, and submitted it to bulk post-polymerization in a TGA chamber serving as a micro-reactor. Bulk post-polymerization was conducted under nitrogen flow (10 mL·min^{-1}) and resulted in a $\Delta\overline{M_w}$ of 180%, with the achieved $\overline{M_w}$ being 5600 g·mol^{-1} and the initial ca. 2000 g·mol^{-1}, Table 1) In both works, the enzymatically synthesized products were successfully upgraded while maintaining their truly green character, as the applied upgrade methods are solventless, not requiring catalysts, conducted in low temperatures, namely fully sustainable.

Thanks to its high quality and purity, an upgraded enzymatically synthesized PBS grade could be a promising candidate carrier in controlled release systems, including active food packaging applications. Conventionally synthesized PBS, poly(lactic acid) (PLA), cellulose acetate phthalate (CAP), poly(ε-caprolactone) (PCL) and poly (D, L-lactide-*co*-glycolide) (PLGA) have already been tested for this purpose [31–34]. Still, to the best of our knowledge, an enzymatically synthesized PBS grade has never been examined for controlled release systems in the open literature. The mostly encapsulated active compounds are antioxidants, including polyphenol bioactive agents such as curcumin, essential oils and resveratrol, since oxidation is one of the main causes of food spoilage. Antioxidants, when used in food packaging, prevent oxidation. Detrimental effects such as vitamin degradation and loss of essential fatty acids, which decrease the nutritional value of the food, are also prevented [35]. It is important to mention, though, that every upcoming technology must be critically evaluated and constantly improved. For example, there may be safety issues such as the package's components (i.e., (bio)polymer and/or other additives) migration to the food, which could risk human health [36]. Another challenge is establishing the optimal concentration distribution of antioxidants around the food to match the release kinetics with the oxidation kinetics of the food.

In this context, PBS was herein enzymatically synthesized *via* a solvent-free process, conducted in two stages to avoid the monomers' loss. The most critical parameters of the process, including reaction

temperature, pressure, and time, were studied in terms of molecular weight, thermal properties, and morphology of the prepolymer, always aiming at the lowest energy requirements, in line with the green chemistry guidelines. Due to the solvent-free process's simplicity, scaling up was achieved based on the optimized conditions resulting in 20 g of enzymatically synthesized PBS for the first time, thus filling the relevant gap in the open literature; in the majority of the relevant published works dealing with PBS enzymatic polymerization, the synthesized polymer's quantity is significantly lower (i.e., about 1–3 g). The non-scalability of the abovementioned processes is probably due to the requirement of various solvents and/or the very high vacuum, even for the small scales, considering that mass transfer limitations (e.g., higher by-product removal rate could be required) are expected to be more intense on large scales. Following the enzymatic prepolymerization, different sustainable post-polymerization methods, such as SSP and melt-post polymerization, were employed to increase the molecular weight of the scaled-up PBS prepolymer. Combining a sustainable post-polymerization method, such as SSP and/or melt post-polymerization, with enzymatic prepolymerization is a novel, alternative approach whose investigation started on previous studies of the authors' group, and the results have been found more than encouraging [6,27]. In our work, pure PBS oligomers were prepared through a simple and scalable process, making them valuable for applications where characteristics, including controlled molecular weight and crystallinity, can be more important than high molecular weights. Such applications can be found in biomedical materials and biodegradable plastics for packaging. The upgraded PBS was then examined for the first time as a carrier for polymeric encapsulation systems *via* emulsification-solvent evaporation technique. Using a high-quality and -purity enzymatically synthesized and upgraded PBS as a carrier in encapsulation systems is a new, cost-effective, green approach that could be useful. Naringin, a plant secondary metabolite, is a flavonoid with known antioxidant activity, thus it was selected as the active compound for encapsulation and preliminary release experiments were conducted to roughly assess the system's relevant profile.

2. Experimental

2.1. Materials

Diethyl succinate (DES, purity 98%) and 1,4-butanediol (BDO, purity 99%) were purchased from Alfa Aesar (Germany). Chloroform (HPLC grade, purity ≥ 99.8%) was purchased from Fisher Scientific (U.K.). The used biocatalyst was C. *Antarctica* Lipase B (CALB) physically absorbed within the macroporous resin Lewatit VPOC 1600 (Novozym 435, N435). Naringin (MW 580 g·mol^{-1}) and poly(vinyl alcohol) (PVA, MW 72,000 g·mol^{-1}, DH 85–89%) were purchased from Tokyo Chemical Industry (Japan) and ITW Reagents (Italy) respectively. Acetone (purity ≥ 99.5%) was purchased from KaloChem (Greece). All chemicals were used without further purification.

2.2. Enzymatic prepolymerization

Stoichiometric amounts of DES (6 mmol, 0.54 g) and BDO (6 mmol, 1.04 g) were premixed in a round-bottom flask and stirred at 40 °C in a rotary evaporator (Rotavapor R-210, Bucchi, Switzerland) under atmospheric pressure. 10% wt N435 (relative to monomers) was subsequently added, and a two-step prepolymerization process was applied. The first step was conducted at 40 °C under atmospheric pressure for 24 h to minimize BDO losses. Then, the temperature was increased and vacuum was applied. At the end of the reaction, chloroform was added (4:1 w/w relative to monomers) to dissolve the formed polyester and the enzyme was subsequently filtered off. The remaining chloroform was evaporated in a high-vacuum pump (Edwards RV5 Rotary Vane Pump, Edwards, U. K.) and the formed polyester (w_{exp}) in the form of a colorless, free-flowing powder was stored in a desiccator. The process yield (%) was calculated by Eq. (1):

$$Process\ yield\ (\%) = \frac{w_{exp}}{w_{theor}} \cdot 100 \quad (1)$$

where w_{exp} is the experimental amount of the product (g), and w_{theor} the theoretical weight of the product predicted by stoichiometric calculations (g).

The variation of enzymatic prepolymerization conditions resulted in three sets of prepolymers (Table 2): samples A (reaction temperature, 80–95 °C), samples B (reduced pressure value, 200 and 20 mbar), samples C (reaction time, 2 and 6 h). The sample B-90 was scaled up to prepare adequate amounts (ca. 10 and 20 g) for the following post-polymerization stage.

Table 2
Conditions of PBS enzymatic prepolymerization and post-polymerization.

Sample	1st step			2nd step			Process yield (%)
	T (°C)	P (mbar)	t (h)	*T* (°C)	P (mbar)	t (h)	
Enzymatic prepolymerization							
A-80	40	1000	24	80	200	2	56
A-85	40	1000	24	85	200	2	84
A-90	40	1000	24	90	200	2	75
A-95	40	1000	24	95	200	2	91
B-90[1]	40	1000	24	90	20	2	81
B-95	40	1000	24	95	20	2	70
C-90	40	1000	24	90	20	6	83
C-95	40	1000	24	95	200	6	94
Post-polymerization							
Precrystallization	90	20	2				
Solid-state post-polymerization							
S8_95[2]	95	20	8				
S2_M2[3]	100	20	2	100	20	2	
S2_M4	100	20	2	100	20	4	
S2_M6	100	20	2	100	20	6	
Melt post-polymerization							
M4_110[4]	110	20	4				

Where S: sample submitted to SSP, M: sample submitted to melt post-polymerization, X: SSP reaction time (h), Y: SSP reaction temperature (°C), Z: melt post-polymerization reaction time (h), W: melt post-polymerization reaction temperature (°C).
[1] B-90_10g and B-90_20g: scaled up prepolymers (10 and 20 g) with process yields 95 and 96% respectively.
[2] SX_Y.
[3] SX_MZ.
[4] MZ_W.

2.3. Post-polymerization

The sample B-90_20g was submitted to SSP and melt post-polymerization. All the post-polymerization processes were conducted in a round-bottom flask in a rotary evaporator (Rotavapor R-210, Bucchi, Switzerland) under reduced pressure (20 mbar). The selected SSP temperature was near the prepolymer's melting point (T_m-T_{SSP}: 7 °C) to achieve increased segmental mobility while maintaining the solid state. The total reaction time was 10 h, including a precrystallization step (2 h) and SSP (Table 2). Melt post-polymerization was conducted at 110 °C, close to the prepolymers melting point, i.e., 12 °C above T_m. At the end of the reactions, the post-polymerized polyester was cooled at room temperature and stored in a desiccator.

2.4. Encapsulation of naringin in polymeric nanoparticles

PBS nanoparticles were prepared according to the emulsification-solvent evaporation technique. For a loading 40% (mass of naringin per mass of polymer used, w·w^{-1}), PBS (300 mg) was dissolved in chloroform (30 mL) and naringin (120 mg) in acetone (9 mL). The organic solutions were mixed and added in an aqueous solution of PVA

(1% w·v^{-1}). The mixture was submitted to sonication (160 Watt, for 6 min). The emulsion was left in an incubator (37 °C, 100 rpm) overnight for solvent evaporation. The nanoparticles were recovered by five runs of centrifugation (20,000 rpm, 20 min, 4 °C) to entirely remove residual PVA. The supernatant from the first centrifugation run was stored at room temperature for indirect quantification of compound encapsulation efficiency. The nanoparticles were finally submitted to freeze-drying and stored in a desiccator. Unloaded (blank) nanoparticles were prepared using the same procedure without the addition of naringin. The process yield was calculated based on Eq. (1), where w_{exp} is the mass of the obtained nanoparticles after freeze-drying (g), and w_{theor} the theoretical weight of the nanoparticles, namely the sum of the initial polymer mass and the initial naringin mass (g).

2.5. Characterization methods

2.5.1. ^{1}H NMR spectroscopy

^{1}H NMR spectroscopy was performed in $CDCl_3$ on a Bruker DRX 400 spectrometer, equipped with a 5 mm ^{1}H/^{13}C dual inverse broad probe operating at 400 MHz to verify the structure of the formed polyesters. The number-average molecular weight ($\overline{M_n}$, g·mol^{-1}) and number-average degree of polymerization ($\overline{x_n}$) were also calculated by Eqs. (2) and (3) respectively [37]:

$$\overline{M}_n = \frac{\frac{I_a - I_d - \frac{2}{3}I_e}{4}.88.10 + \frac{I_c}{4}.84.07 + \frac{I_d}{2}.89.11 + \frac{I_e}{3}.45.06}{0.5.\left(\frac{I_d}{2} + \frac{I_e}{3}\right)} \quad (2)$$

where 88.10 is the molar mass (g·mol^{-1}) of 1,4-butyl segments, 84.07 is the molar mass (g·mol^{-1}) of succinate segments, 89.11 is the molar mass (g·mol^{-1}) of 1,4-BDO end-groups, 45.06 is the molar mass (g·mol^{-1}) of ester end-groups and I_a is the integral of the 1,4 butyl segment signal at 4.12 ppm, I_c is the integral of the succinate segment signal at 2.62 ppm, I_d is the integral of the BDO end-group signal at 3.67 ppm and I_e is the integral of the ester end-group signal at 1.25 ppm.

$$\overline{x_n} = 2.\frac{\overline{M}_n}{m_0} \quad (3)$$

where m_0 is the molecular weight of the repeating PBS unit (172 g·mol^{-1}).

The percentage of the ester-end group in respect to the total number of prepolymer ends was calculated by Eq. (4):

$$ester\ end\ groups\ percentage\ (\%) = \frac{\frac{I_e}{3}}{\frac{I_e}{3} + \frac{I_d}{2}}.100 \quad (4)$$

2.5.2. Fourier transform infrared spectroscopy (FTIR)

FTIR was performed on an Alpha II (Alpha II, Bruker, Germany) using the ATR method with a diamond crystal in the range of 400–4000 cm^{-1} wavenumber region and a resolution 4 cm^{-1}. The prepolymers' being in the form of powder, were submitted to the FTIR analysis without prior processing.

2.5.3. Gel permeation chromatography (GPC)

GPC was carried out with the use of Agilent 1260 Infinity II instrument (Agilent Technologies, Germany), equipped with a guard column (PLgel 5 μm) and two PLgel MIXED-D 5 μm columns (300 × 7.5 mm). Elution was carried out with chloroform (≥99.8% purity, Fisher Scientific, U.K.) at a flow rate of 1 mL·min^{-1}. The analysis was performed using an Agilent 1260 Infinity II refractive index detector (RID) (G7162A). The calibration of the instrument was carried out with polystyrene standards of molecular weight from 162 to 500,000 g·mol^{-1} (EasiVial PS-M 2 mL, Great Britain) and a universal calibration curve was constructed.

2.5.4. Differential scanning calorimetry (DSC)

Differential scanning calorimetry (DSC) measurements were performed in a Mettler DSC 1 STARe System. Heating – cooling – heating cycles from 30 to 150 °C, cooling to − 10 °C and finally heating to 150 °C, were conducted for the enzymatically synthesized and post-polymerized PBS grades, and heating from 30 to 250 °C was conducted for the (loaded and unloaded) PBS nanoparticles. All measurements were conducted under N_2 flow (20 mL·min^{-1}), with a rate of 10 °C·min^{-1}. The melting points derived from the first and second heating cycle are represented as T_{m1} and T_{m2}, and the relevant mass fraction crystallinity (x_c, %) was computed according to Eq. (5). The melt crystallization point (T_c), as well as the crystallization enthalpy (ΔH_c, J·g^{-1}) were obtained from the DSC cooling cycle.

$$x_c(\%) = \frac{\Delta H_f - \Delta H_{cc}}{\Delta H_0}.100 \quad (5)$$

where ΔH_f is the heat of fusion (J·g^{-1}), ΔH_{cc} is the cold crystallization enthalpy (J·g^{-1}), ΔH_0 is the heat of fusion of 100% crystalline PBS (J·g^{-1}) equal to 110.5 J·g^{-1} [38].

2.5.5. Thermogravimetric analysis (TGA)

Thermogravimetric analysis (TGA) was conducted in a Mettler TGA/DSC 1 thermobalance from 30 to 600 °C, at a heating rate 10 °C·min^{-1} under N_2 flow (10 mL·min^{-1}). The onset decomposition temperature was defined as the temperature at 5% weight loss ($T_{d,5\%}$), the degradation temperature (T_d) was determined at the maximum rate of weight loss, and the char yield as the % residue at 600 °C.

2.5.6. Encapsulation efficiency determination

The encapsulation efficiency (EE%) was determined using UV–Vis spectroscopy. For that purpose, the supernatant from the first centrifugation run, during the preparation of loaded nanoparticles, was decanted and the naringin content was determined using UV–Vis spectroscopy (λmax 280 nm).

2.5.7. Dynamic Light Scattering (DLS)

Mean particle size, polydispersity index (PDI) and ζ-potential of loaded and blank nanoparticles were determined *via* Dynamic Light Scattering (DLS) technique using a Zetasizer Nano ZS (Malvern Instruments, UK). The samples were prepared by dispersing 0.2 mL of nanoparticle suspension in water (4 mL), resulting in off-white opaque aqueous suspensions. All measurements were performed in triplicate at 25 ± 1 °C.

2.5.8. In vitro release study

Preliminary *in vitro* release of naringin from the polymeric nanoparticles was conducted using the dialysis method. Experiments were conducted by dispersing 26 mg of loaded nanoparticles into 5 mL of water and adding the dispersions in SEVAPOR® membranes (pore diameter approx. 25 Å). The membranes were immersed in beakers containing 40 mL phosphate buffer (pH 7.4) under magnetic stirring at 25 °C ± 1 °C. The amount of naringin released in each beaker was determined using UV–Vis spectroscopy.

3. Results and discussion

3.1. Enzymatic prepolymerization

3.1.1. Effect of reaction temperature

The herein applied process was conducted in two steps to minimize the monomers' loss, especially due to the BDO's high volatility. The selected reaction temperature of the first stage was 40 °C, based on a previous study of the authors' group, where BDO and DES's enzymatic polycondensation was studied in solvent-based (low boiling point solvents, namely isooctane and toluene) and solvent-free systems at three

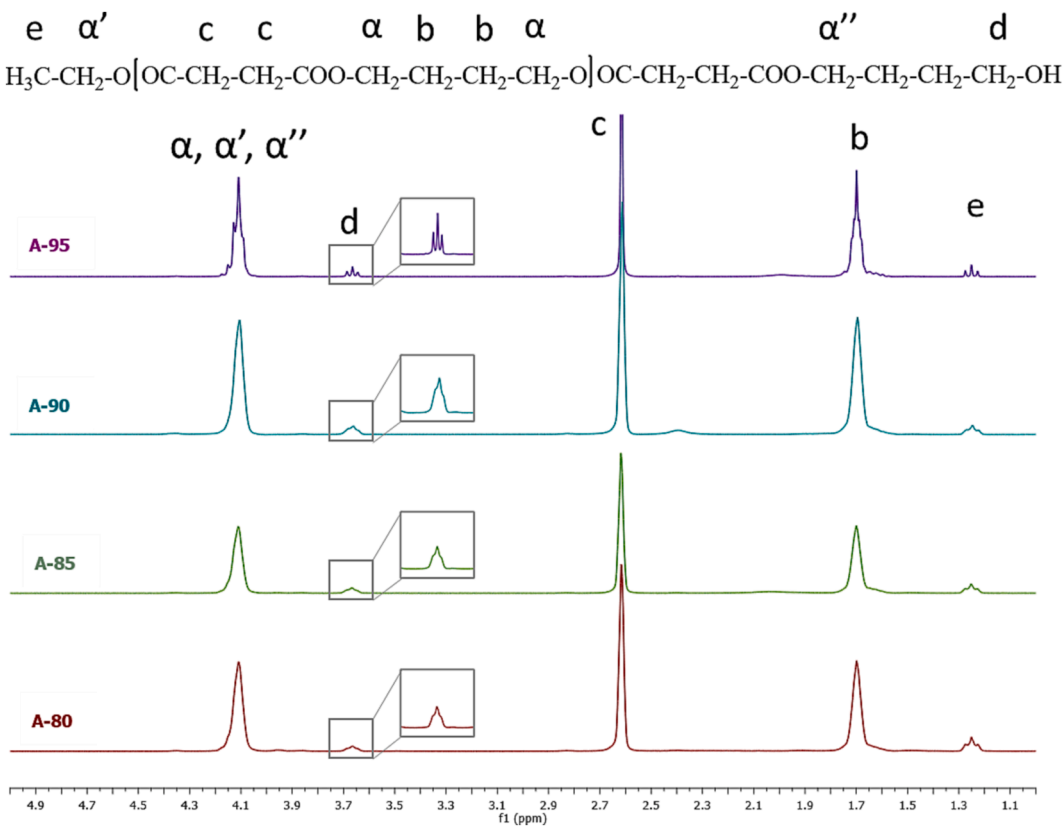

Fig. 1. ^{1}HNMR spectra of the prepolymers A-80, A-85, A-90 and A-95.

reaction temperatures (40, 50 and 60 °C) in the presence of N435 [6]. The bulk system was found more effective than the examined solvent-based systems; this is attributed to the fact that a polar solvent may deactivate the enzyme due to conformational changes and be responsible for releasing water bound to the enzyme leading to hydrolysis reactions. Thus, the PBS prepolymer synthesized in bulk at 40 °C presented the highest $\overline{M_n}$ (2800 g·mol^{-1}), probably indicating increased enzyme activity at this temperature.

The second stage's reaction temperature (80–95 °C, samples A) was herein investigated for the enzymatic polymerization of PBS. The relevant process yields were in the range of 56–91%, with the lowest and the highest value for 80 °C and 95 °C, respectively (Table 2). For all samples A, PBS repeating unit was verified *via* ^{1}H NMR spectroscopy (Fig. 1). The determined shifts were: 4.11–4.18 ppm (4H, —O—C***H₂***—CH₂—CH₂—C***H₂***—O— from the 1,4-butyl segment of BDO), 2.62 ppm (4H, —CO—C***H₂***—C***H₂***—CO—, from the succinate segment of DES), 1.65–1.75 ppm (4H, O—CH₂—C***H₂***—C***H₂***—CH₂—O, from the1,4-butyl segment of BDO), 3.65–3.69 ppm (2H, HO—C***H₂***—CH₂—CH₂—CH₂—O, from the BDO end-groups) and 1.23–1.28 ppm (3H, C***H₃***—CH₂—O—CO—, from the ester end-groups). The BDO end-group HO—CH₂—CH₂—CH₂—C***H₂***—O (α'') and the ester end-group CH₃—C***H₂***—O—CO— (α′) both give a signal at 4.12 ppm which is overlapped with the main butylene signal (—O—C***H₂***—CH₂—CH₂—C***H₂***—O—) of the repeating unit.

Except for the PBS repeating unit verification, the ^{1}H NMR spectra indicated no residual BDO in the final products. The hydroxyl end group peak (signal at about 3 ppm [39]) monitored in the BDO spectrum, is absent in all prepolymers spectra (Fig. S1). The resonance signal of the end unit (—OH) proton is typically not detectable in polymers' spectra due to intra-molecular hydrogen bonding [40] (e.g., between hydroxyl groups and the ester bonds). It can be thus considered that there is no residual unreacted BDO in the prepolymers.

A-95 spectrum presented a typical morphology, with multiple,

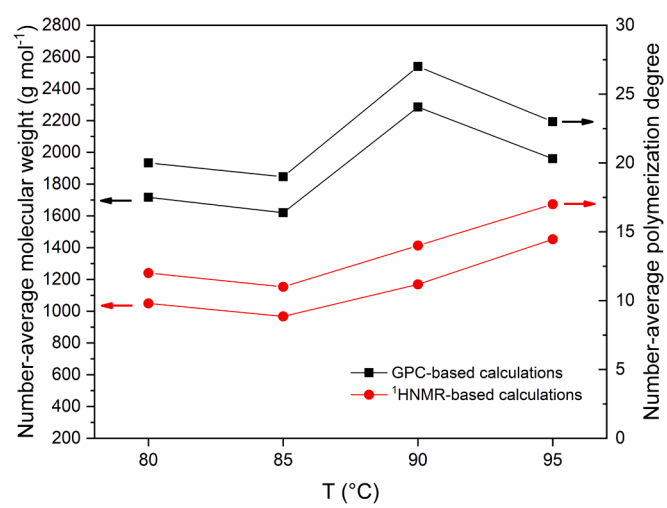

Fig. 2. ^{1}HNMR- and GPC-calculated $\overline{M_n}$ and $\overline{X_n}$ of the A-80, A-85, A-90 and A-95 prepolymers.

distinguished peaks at 4.12 and 1.7 ppm [6,37]. On the contrary, the A-80, A-85, and A-90 spectra presented single, unsplit peaks at 4.12 and 1.7 ppm. This difference indicates decreased chain length of the A-80, A-85 and A-90 prepolymers compared to A-95, given that there is a minimum polymerization degree above which the ^{1}HNMR peaks distinguish [41]. The ^{1}HNMR-based molecular weight calculations (Fig. 2) verified this observation, since decreased values were found for the A-80, A-85 and A-90 prepolymers compared to A-95. Based on these calculated results, for $\overline{M_n}$ and $\overline{X_n}$ values lower than 1200 g·mol^{-1} and 14 respectively (A-90 prepolymer), the monitored peaks in the spectrum do not distinguish. A similar trend has been observed in previous results of the

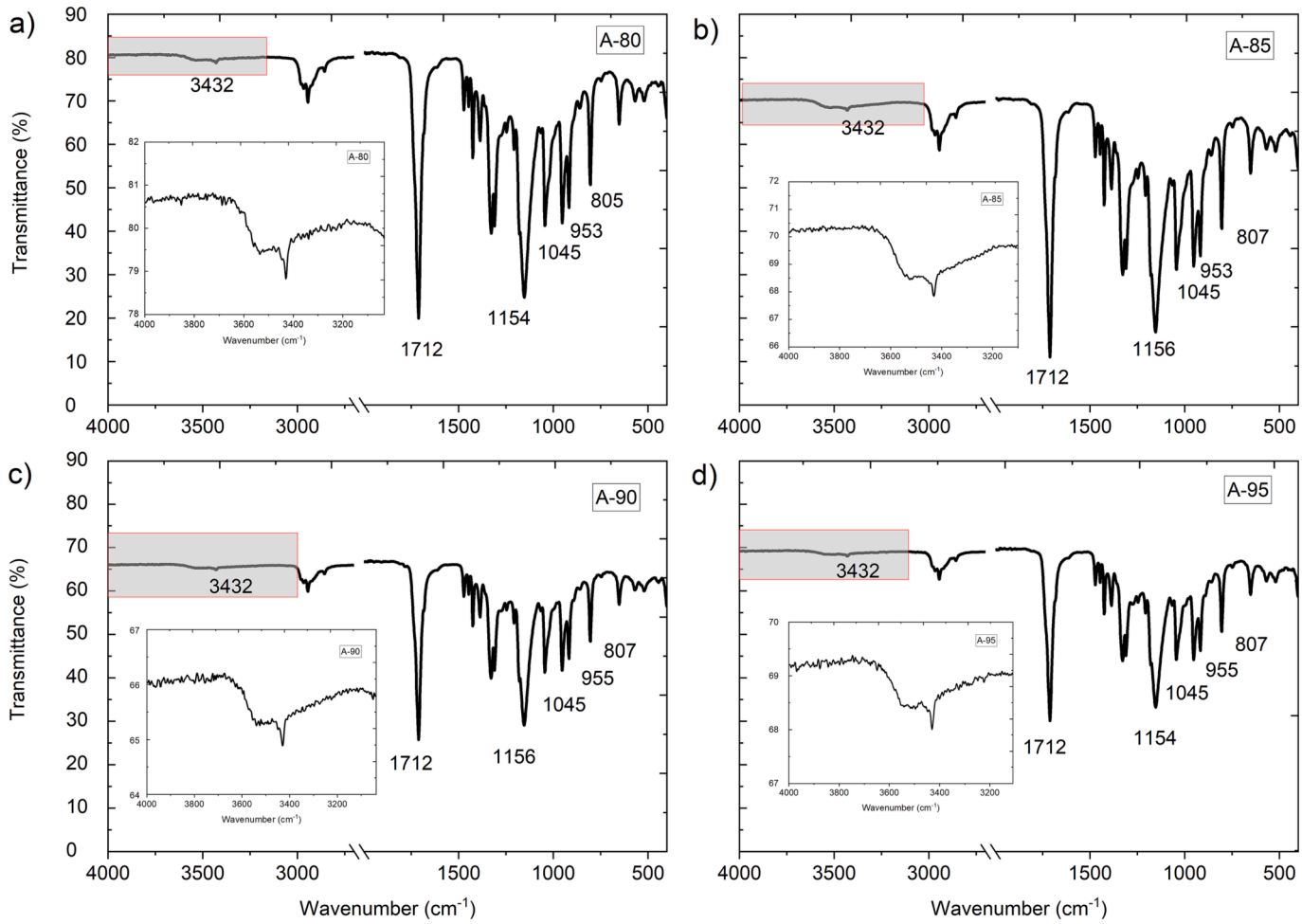

Fig. 3. FTIR spectra of the prepolymers A-80 (a), A-85 (b), A-90 (c) and A-95 (d).

authors' group; higher molecular weight PBS oligomers (e.g., $\overline{M_n}$ 2000 g·mol^{-1}) presented spectra with clearly separated peaks at 4.12 and 1.7 ppm in contrast to the spectrum of lower molecular weight PBS oligomers (e.g., $\overline{M_n}$ 1250 g·mol^{-1}), where the same peaks were slightly separated. Similarly, Kang-Jen Liu [41] used oligomers (n = 3–10) and high molecular weight grades of poly(dimethyl siloxane) (PDMS) to prove the strong "short-range near-neighbor effect" on the NMR spectra as a function of chain length in bulk and solution. The researcher interestingly observed that the "internal" methyl groups of the polymer were distinguishable only when the chain length increased to n ≥ 5. The main polymer NMR peaks of the polymers were also affected by chain length when aromatic solvent media were used. It was concluded that these phenomena should be considered, especially when low molecular weight polymer samples are measured.

Interestingly, the A-prepolymers $\overline{M_n}$ values defined *via* GPC significantly differed (33–92%) from the relevant ^{1}HNMR-derived values (Fig. 2, Table S1). This phenomenon is already observed in the literature. Adams et al. [42] used ^{15}N NMR and GPC to calculate the molecular weight of polyaniline samples. The GPC-calculated molecular weight values were found artificially higher due to either incomplete polyaniline chain disentanglement in solution or the polyaniline's stiffer nature compared to the used standard (PVP). Da Costa et al. [43] used four PBS samples of different molecular weights to compare the ^{1}HNMR- and the GPC-calculated molecular weight values. The values presented significant deviations (3–100%). It was concluded that NMR presents absolute, more accurate results than GPC. Similarly, Izunobi and Higginbotham [44] assessed the molecular weight values of the homopolymer α-methoxy-ω-aminopolyethylene glycol (MPEG-NH2) and the block copolymer α-methoxy-polyethylene glycol-block-poly-ε-(benzyloxycarbonyl)-L-lysine (MPEG-b- PLL(Z)) *via* different methods including GPC (calibrated with PS and PEG standards) and ^{1}HNMR. The strong dependence of the GPC results on the calibrant was proved as the $\overline{M_n}$ values obtained from the PS-calibrated curve were lower than the theoretical molecular weights of the measured samples, whereas the opposite scenario occurred for the PEG standards. In agreement with the abovementioned published works, the researchers concluded that ^{1}HNMR is the most reliable technique to assess the molecular weight values, considering that there is no need for calibration. In any case, in our work, the trend of the monitored molecular weights through ^{1}HNMR and GPC is identical for every temperature except for 95 °C. This inconsistency of A-95 could probably be attributed to inhomogeneity of the prepolymer due to partial loss of the enzyme's reactivity at elevated temperatures.

The A-prepolymers chemical structure was also verified *via* PBS FTIR characteristic peaks (Fig. 3): 1712 cm^{-1}, C═O stretching vibrations; 1154 cm^{-1}, C—O—C stretching vibration from the ester groups; 1045 cm^{-1}, O-C—C vibration in the BDO segment; 953 cm^{-1}, C—O symmetric stretching mode and 805 cm^{-1}, CH_2 in the succinate in-plane bending mode [37,45]. The broad band at around 3400 cm^{-1} is attributed to the O—H stretch in the BDO end-group. The intensity of the carbonyl stretch C═O band (1712 cm^{-1}) is proportional to the concentration of the ester bonds (COO) along the polymer backbone and at the end of the polymer chain. In this work, where the synthesized PBS grades were oligomers, the contribution of the ester end-groups to the overall carbonyl stretch C═O band signal intensity is expected to be significant.

Indeed, the carbonyl content, defined qualitatively based on the

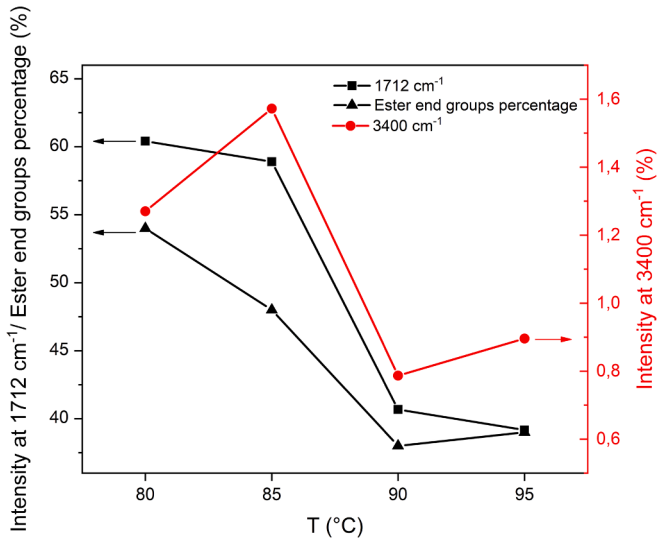

Fig. 4. ^{1}HNMR-calculated ester end groups percentage (Eq. (4)) and peak intensities (transmittance percentage) at 1712 and 3400 cm^{-1} for the prepolymers A-80, A-85, A-90, A-95.

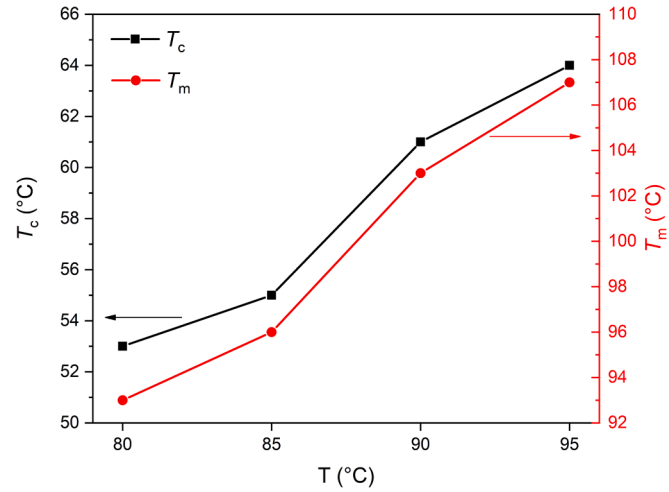

Fig. 5. Crystallization temperature (T_c) and melting point from the second heating (T_{m2}) of the prepolymers A-80, A-85, A-90 and A-95.

intensity at 1712 cm^{-1}, was found to decrease with temperature (Fig. 4), showing that the peak intensity is mainly related to the chain-end ester groups consumed with polymerization. The same trend is also observed for the ^{1}HNMR-calculated ester end groups percentage, which was also found to decrease (Eq. (4), Fig. 4). More precisely, the COO peak intensity and the ester end groups percentage decreased slightly (3 and 11%, respectively) from 80 to 85 °C, indicating that the reaction temperatures below 85 °C are insufficient to reach effective polymerization rates. On the other hand, the highest percentage decrease of the carbonyl content (30%) and the highest decrease of the ^{1}HNMR-calculated ester end groups percentage (21%) were detected above 85 °C, indicating that increased polymerization rates and chain-end ester groups consumption were achieved. The kinetic energy of the enzyme and the substrate chains was sufficient, and the most successful collisions due to higher diffusion rates between the enzyme's active site and the substrates were favored at 90 °C. Further increase of the reaction temperature (from 90 to 95 °C) did not have a significantly positive effect on the polymerization rate, as the carbonyl content further decreased slightly by 4%, while the ^{1}HNMR-calculated ester end groups percentage remained almost constant.

In agreement with the decrease of COO intensity with temperature, the obtained GPC-calculated average molecular weights significantly increased for reaction temperatures above 85 °C (Fig. 2, Table S1). In particular, the highest GPC-calculated $\overline{M_n}$ (2300 g·mol^{-1}) and $\overline{M_w}$ (5100 g·mol^{-1}) were observed at 90 °C with a dispersity of 2.2, being close to the most probable molecular weight distribution value. At 95 °C, the average molecular weights seemed to decrease (by 15%) compared to 90 °C remaining though higher than the values at 80 and 85 °C. This behavior of lower polymerization rate at 95 °C may be due to partial loss of the enzyme's reactivity, while dispersity remained at similar values (2.1). A recent paper correlated this effect of biocatalysis temperature to the melting of the formed PBS chains above 84 °C [24]. More precisely, based on the DSC results, the solid state of PBS below 83.6 °C limited the diffusion of polymer chains through the porous support to reach the enzyme's active site. According to the researchers, PBS melting was possible above this temperature, and 90 °C was thus found as the optimal reaction temperature. Similarly, 90 °C has been indicated as the optimal reaction temperature of N435 in different polymerization systems (e.g., ring-opening polymerization of ε-caprolactone in toluene [46] and furanic-aliphatic polyamides synthesis in toluene [47]).

The intensity of the hydroxyl end-groups peak at 3400 cm^{-1}, which is much weaker and broader than the COO band at 1712 cm^{-1}, was also examined for the A-prepolymers (Fig. 4). The most intense change was detected when the reaction temperature increased from 85 to 90 °C (50% decrease), similar to the carbonyl content. It is thus confirmed that above 85 °C, increased transesterification rates were achieved. At the lower temperatures of 80 and 85 °C, the transesterification rates were significantly lower, and OH groups appeared to increase at 85 °C compared to 80 °C. This OH-end group concentration increase may indicate the initial formation of OH-rich oligomers during lipase-catalyzed polymerization: due to sterical factors, increased diol concentration can be foreseen in the enzyme's active site, given that the BDO carbon content is lower than DES [6,18].

Turning to the thermal properties (Fig. S2, Table S2), all the enzymatically synthesized samples were submitted to a first heating cycle to erase their thermal history, which was followed by cooling up to −10 °C. The A-prepolymers crystallization temperatures (T_c) were correlated to prepolymerization temperature and thus to the achieved molecular weights (Fig. 5). The higher molecular weight prepolymers (A-90 and A-95) presented higher crystallization temperatures (61 and 64 °C respectively) in contrast to the lower molecular weight A-80 (T_c = 53 °C) and A-85 (T_c = 55 °C). This trend is attributed to the increased ease of chain folding of longer macromolecules and the decreased number of end groups that may act as crystal defects slowing down the crystallization process. During the second heating, cold crystallization occurred for all the A-prepolymers at 68–81 °C, indicating the formation of small crystals that didn't have enough time to be formed during cooling from the melt. The higher molecular weight prepolymers (A-90 and A-95) presented increased ΔH_{cc} values (8 and 12 J·g^{-1} respectively), confirming again their enhanced tendency to crystallize. Multiple melting peaks were then monitored for all the A-prepolymers, indicating the existence of different crystal populations (e.g., different lamellar thickness) with different melting stability, a typical phenomenon for PBS. The first two melting peaks (77–92 and 84–100 °C respectively) are attributed to the original crystalline melting [48], while the third one, monitored at the highest temperatures (93–107 °C) corresponds to the recrystallization of the partially melted crystals during heating and was found close to the typical T_m (114 °C) of a commercial PBS grade [45]. In all the prepolymers' DSC thermograms, no thermal transitions related to BDO (T_m 16 °C) were monitored (Fig. S3a), indicating the absence of residual BDO in the final products, in agreement with the above-mentioned ^{1}H NMR results. As expected, the higher molecular weight prepolymers (A-90 and A-95) presented the higher melting points (Fig. 5) and sharper endothermic peaks (Fig. S2c) similarly to typical PBS thermographs. The melting enthalpies were also found increased for the higher molecular weight grades A-90 and A-95, i.e., 88 and 86 J·g^{-1}

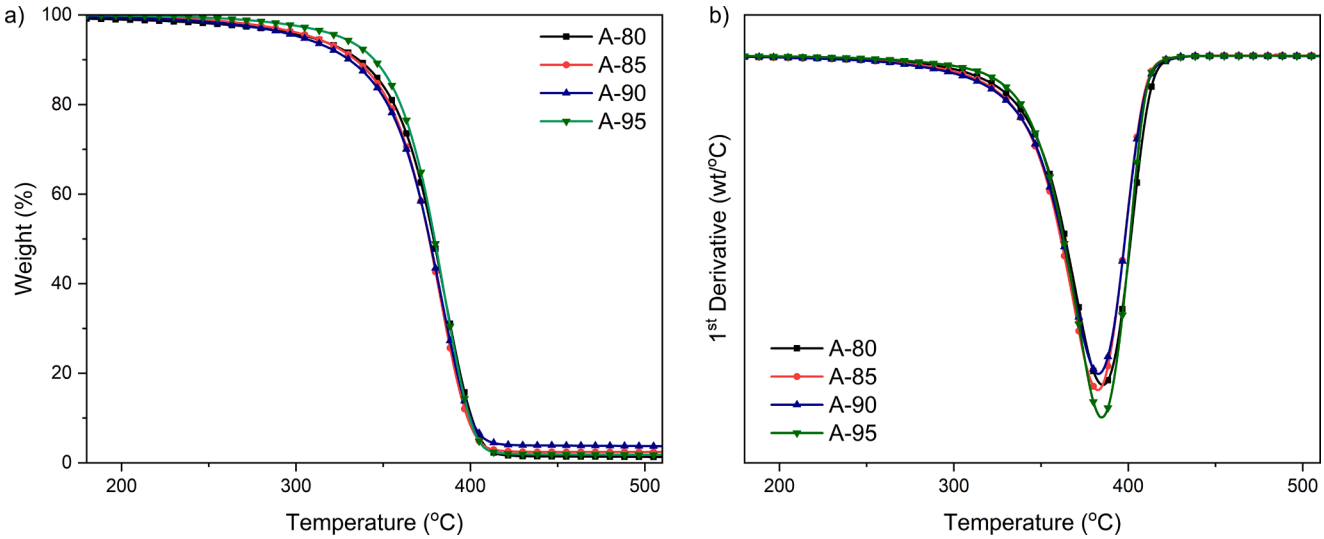

Fig. 6. TGA curves and 1st derivatives of the prepolymers A-80, A-85, A-90 and A-95.

respectively, corresponding to x_c of 73 and 67%.

Turning to TGA results, all A-prepolymers presented a single-step decomposition profile (Fig. 6), with $T_{d5\%}$ and T_d from 303 to 326 °C and 382 to 398 °C (Table S3). The monitored T_d values were found similar to a commercial PBS grade (T_d at ca. 400 °C) [45], and no mass loss was monitored at temperatures related to the monomers evaporation (T_b of BDO: 205 °C, T_b of DES: 179 °C) indicating also no residual monomers in the products (Fig. S3b). The achieved high thermal stability of the enzymatically synthesized PBS grades, despite their low molecular weight, can be evidenced and attributed to the enzymes' selectivity, the applied mild polymerization conditions and the absence of undesirable by-products and metal catalyst residues that may accelerate thermal degradation reactions lowering thus the T_d.

3.1.2. *Effect of reduced pressure*

Given the improved characteristics of the A-90 and A-95 prepolymers, the reduced pressure effect was studied at 90 and 95 °C in order to enhance the rate of by-product (ethanol) removal *via* evaporation and diffusion. In samples B-90 and B-95, the pressure was set at 20 mbar, and the reference prepolymers were A-90 and A-95 (200 mbar).

Concerning the macroscopic observations, B-90 was a colorless solid in contrast to B-95, which was in the form of a sticky bulk material, indicating a lower degree of polymerization. The process yield slightly increased at 90 °C (81%) compared to A-90 (75%), while it significantly decreased at 95 °C (Table 2): from 91% (A-95) to 70% for B-95. At the end of the reaction at 95 °C under 20 mbar, the enzyme presented yellowish color (Fig. S4a), indicating the beginning of its denaturation, which probably led to the formation of lower molecular weight oligomers that are more susceptible to sublimation, justifying also the lower yield.

The chemical structure of the prepolymers was again verified *via* ^{1}HNMR, and the molecular weights were calculated *via* ^{1}HNMR and GPC (Table S1). According to the process yield values, GPC-calculated $\overline{M_n}$ slightly increased with the reduced pressure for the prepolymers synthesized at 90 °C. On the other hand, B-95 presented significantly lower molecular weight values, and GPC showed a narrower MWD (dispersity 1.5), something that may also confirm the removal of the lower molecular weight oligomers through sublimation under the high vacuum of 20 mbar. Given that the firstly formed low MW oligomers are the most susceptible to sublimation, the enzyme was constantly submitted to start

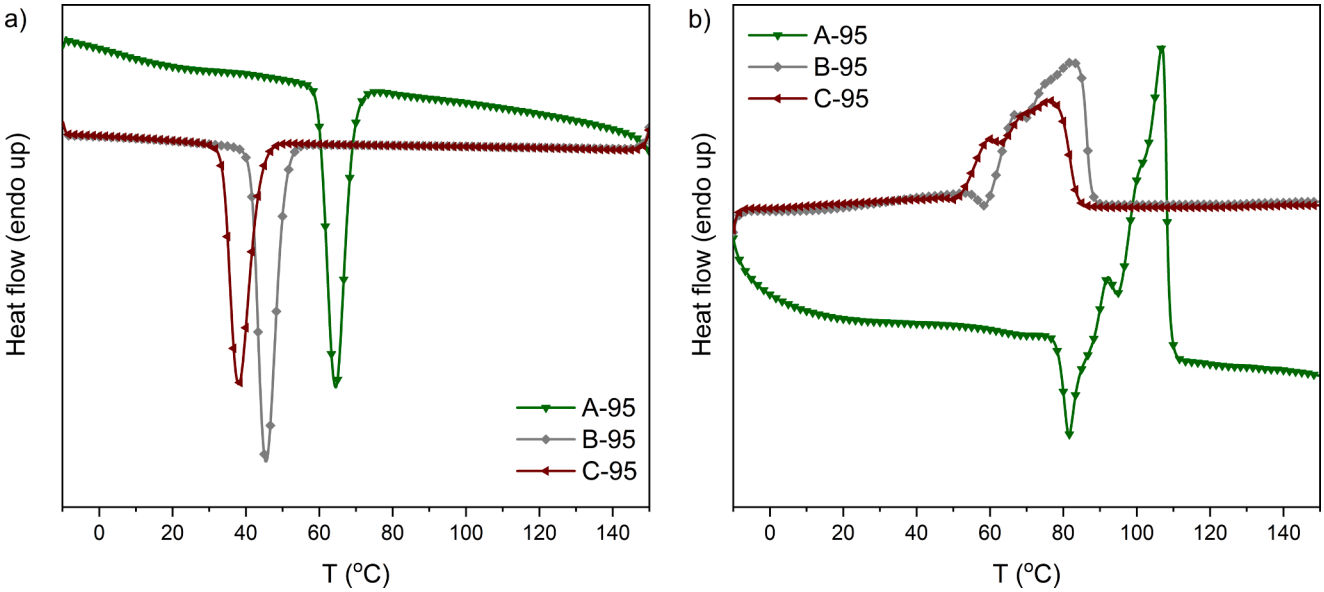

Fig. 7. Cooling (a) and second heating (b) of the prepolymers A-95, B-95 and C-95.

new catalytic cycles, as the reaction propagation was impossible. The accelerated enzyme's deactivation compared to A-95 can be thus attributed to the high complexity of each catalytic cycle, including multiple stages (e.g., intermediates' formation and stabilization, nucleophile attacks, release of the product) [18].

As regards DSC analysis, A-90 and B-90 behaved similarly (Table S2). Melt crystallization (T_c) occurred at ca. 61–64 °C, and the presence of three melting peaks, all slightly higher for B-90 (91, 99 and 105 °C) is in line with the slightly increased molecular weight of B-90 compared to A-90. On the other hand, the prepolymers synthesized at 95 °C, presented significantly different thermal properties. B-95 had significantly decreased T_c (45 °C) compared to A-95 (64 °C), and similar ΔH_c 81 and 84 J·g^{-1} (Fig. 7). The faster crystallization kinetics of A-95 verified its higher molecular weight compared to B-95. Such a trend has been observed in different PBS [49] and poly(butylene sebacate) [50] grades with $\overline{M_w}$ of 4600–18,000 g/mol and 12,000–16,000 g/mol respectively. More precisely, as MW was increasing, the T_c values also increased by 10 °C for PBS (77–87 °C) and by 4–5 °C for poly(butylene sebacate) (56–60 °C) indicating the increased tendency of the higher molecular weight grades compared to the lower molecular weight probably due to the increased lamellar crystal thickness derived from the increased ease of chain folding of the longer macromolecules. For the same reason, during the second heating, B-95 presented significantly broader endothermic peaks and severely lower T_m compared to A-95. Indicatively, the third melting endotherm was found 107 °C and 83 °C for A-95 and B-95 respectively. The ΔH_m values of B-95 were also found decreased (72 J·g^{-1}) compared to A-95 (86 J·g^{-1}).

The B-90 prepolymer also presented increased $T_{d,5\%}$ (327 °C, Table S3) compared to the A-90 (303 °C), while the B-95 presented significantly decreased $T_{d,5\%}$ (272 °C) compared to A-95 (326 °C). The $T_{d,5\%}$ is highly sensitive and correlated to the molecular weight changes; the simultaneous increase of reaction temperature and vacuum in B-95 had a negative effect on the achieved molecular weight and on the prepolymer thermal stability. On the other hand, at the temperature of 90 °C, the higher vacuum resulted in a slightly higher overall quality of the prepolymer, i.e., in terms of molecular weight, melting characteristics and thermal stability.

3.1.3. Effect of the reaction time

Given the positive effect of the reduced pressure at 90 °C and its negative effect at 95 °C, a higher reaction time (6 h) was also studied as parameter at 90 °C, 20 mbar (C-90) and 95 °C, 200 mbar (C-95). On that basis, the reference prepolymers were B-90 and A-95 respectively.

Regarding prepolymers morphology, C-90 was a colorless solid (Fig. S4b) similar to B-90. On the contrary C-95 was sticky (Fig. S4c), indicating a low degree of polymerization. Additionally, the enzyme collected at the end of the reaction at 95 °C presented yellow color, an indication of its thermal denaturation (Fig. S4d). The process yield increased slightly for C-90 and C-95 (Table 2) with the increase in the reaction time. For C-90, $\overline{M_n}$ increased up to 3100 g·mol^{-1} being close to the B-90 value (2500 g·mol^{-1}). However, $\overline{M_w}$ was found to decrease, indicating the existence of competitive phenomena. The prolonged reaction time at the optimum temperature of 90 °C is expected to result in the formation of longer PBS chains but also increased by-product amounts requiring a higher vacuum to be removed. As a result, ethanol accumulation within the reacting particles is highly probable, rendering the formed oligomers susceptible to downgrade. Considering also the solid state character of the products at 90 °C, it is easier for the shorter macromolecules to be recombined *via* diffusion and reaction on the enzymatic support, compared to the longer macromolecules which present decreased mobility being thus more susceptible to degradation. As a result, $\overline{M_n}$ of C-90 slightly increased in contrast to $\overline{M_w}$, which was found to decrease due to the longer macromolecules' higher contribution to the weight-average molecular weight calculation. For C-95 °C, both molecular weight averages significantly decreased, confirming the enzyme's time-dependent thermal inactivation at 95 °C [51] indicated by its yellow color at the end of the reaction.

B-90 and C-90 presented similar thermal properties as evidenced by DSC and TGA analyses (Tables S2 and S3). On the contrary, C-95 presented significant differences to A-95, such as slower crystallization kinetics (26 °C lower T_c, 8 J·g^{-1} lower ΔH_c). The increased ΔH_c of the higher molecular weight prepolymer A-95 confirms its enhanced tendency to crystallize compared to the lower molecular weight C-95, attributed to the increased lamellar crystal thickness derived from the increased ease of chain folding of the longer macromolecules. Song and Sung [49] who studied the crystallization of different PBS grades as a function of molecular weight (4600–27,000 g·mol^{-1}) also noticed a significant increase at the ΔH_c (from 18.1 to 19.7 cal·g^{-1}) when the molecular weight increased from 4600 to 6300 g·mol^{-1}. Then ΔH_c remained almost constant up to 11,000 g·mol^{-1} and finally decreased at higher molecular weight values (18.7 cal·g^{-1} at 27,000 g·mol^{-1}). The fact that the higher molecular weight grade (27,000 g·mol^{-1}) presented lower T_c and ΔH_c values, can be attributed to the limited ease of chain folding due to the increased viscosity of the medium. C-95 also presented significantly different melting points, crystallinity and $T_{d,5\%}$, all attributed to its lower molecular weight (Fig. 7, Table S3). Interestingly, the T_d of the prepolymers C-90 (383 °C) and C-95 (383 °C) were found similar to the B-90 (384 °C) and A-95 (385 °C), respectively. Even though C-90 and B-90 presented similar MW values, while C-95 presented a significantly decreased MW compared to A-95, the T_d in both cases was not affected and the decomposition process was conducted in a single-stage. This non-dependence of the T_d and the MW is attributed to the fact that the $\overline{M_n}$ of all the prepolymers is higher than 1000 g·mol^{-1}. More precisely, in a previous work of the authors [6], it was proved that the enzymatically synthesized PBS grades with low molecular weights ($\overline{M_n}$ < 1000 g·mol^{-1}) decomposed at temperatures lower than 450 °C and presented a two-stage decomposition in most cases. On the contrary, higher molecular weight grades ($\overline{M_n}$ > 1000 and up to 3000 g·mol^{-1}) decomposed at temperatures higher than 450 °C in a single-stage process. A similar trend has been reported by Chrissafis et al. [52] for poly(ethylene succinate) or PES; it was proved that PES of $\overline{M_n}$ 2300–3000 g·mol^{-1} decomposed at temperatures lower than 200 °C, with a three-stage decomposition process. On the contrary, PES of $\overline{M_n}$ 3560–21,480 g·mol^{-1} presented a clear mass loss stage at temperatures higher than 290 °C. The enhancement mechanism of thermal stability above a specific MW is rather complex and has not been studied in detail in the open literature. Unger et al. [53] investigated the effect of molecular weight on the thermal degradation of poly(ε-caprolactone) or PCL by TGA and FTIR. The researchers discovered that the lower molecular weight PCL grade (10,000 g·mol^{-1}) decomposed in a three-step mechanism in contrast to the higher molecular weight (80,000 g·mol^{-1}) for which a two-step decomposition was monitored. The products formed at each stage were defined through FTIR, and it was found that in the low molecular weight PCL decomposition during the first step, which was not observed for the high molecular weight grade, the monomer ε-caprolactone was formed. It was considered that the depolymerization was conducted via an unzipping mechanism of the PCL chains. This reaction was restricted to low molecular weight PCL because an increased concentration of (hydroxyl and carboxyl) end groups was present. All the other products formed during the second and third decomposition steps were detected in low and high molecular PCL grades. Summing up, there seems to be a critical molecular weight value above which each polymer's decomposition is less (or even not at all) affected by the molecular weight variations.

3.2. Scaling up of the enzymatic prepolymerization

The prepolymers synthesized at 90 °C under 20 mbar for 2 and 6 h (B-90 and C-90) presented the highest performance in terms of molecular weight ($\overline{M_n}$ 2500 and 3100 g·mol^{-1}, $\overline{M_w}$ 6700 and 5200 g·mol^{-1}),

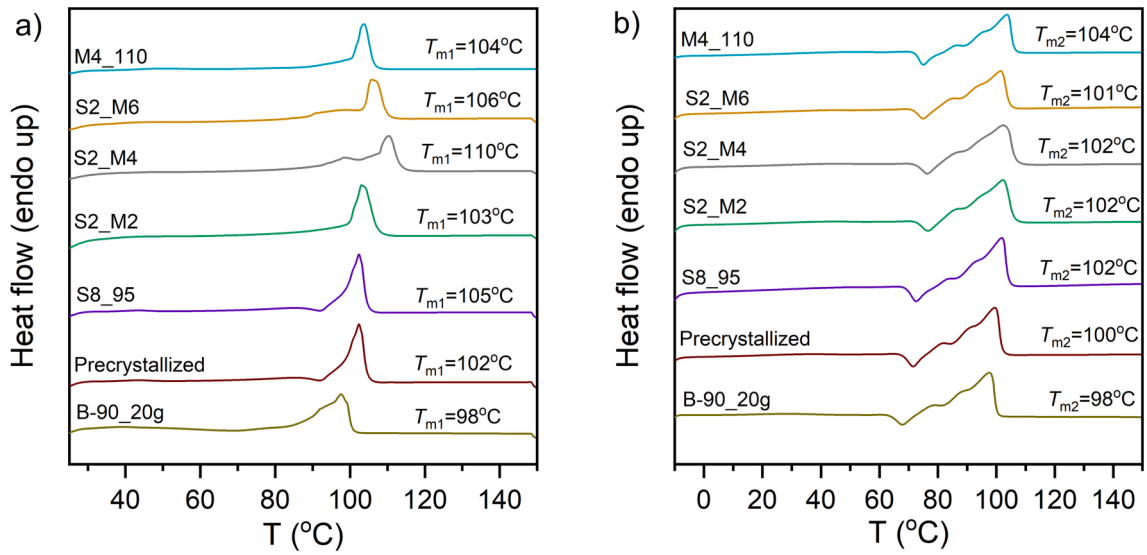

Fig. 8. First heating (a) and second heating (b) of the starting material (B-90_20g) and the post-polymerized samples.

thermal properties (T_{m2} 105 and 107 °C) and morphology (colorless, free-flowing powders). Even though their characteristics were similar, B-90 was selected to be scaled up (10 and 20 g) as its synthesis reaction time was shorter (2 h), permitting thus lower energy requirements in line with the main principles of green chemistry [54]. It is also important that the herein applied process can be easily scaled up, considering that it is a low-temperature bulk polymerization technique not requiring multiple solvents for the final product isolation.

The scaled-up prepolymers were also received as colorless, free-flowing powders, and the achieved process yields were 95–96%. The obtained molecular weight values were found to be lower compared to the smaller scale grades (Table S1); especially the $\overline{M_w}$ presented the highest decrease (53%) from the 1 g- to the 20 g-scale, indicating that the formation of the longer chains was restricted, while the relevant decrease in the $\overline{M_n}$ values was less intense. This phenomenon can be attributed to mass transfer limitations that are more intense in large scales. For instance, the rate of by-product removal decreases as the produced ethanol is increased (constant vacuum of 20 mbar). The scaled-up prepolymers' thermal properties were also lower compared to B-90 (Fig. S5, Table S2). The melting points derived from the first heating of the scaled-up prepolymers, critical for the subsequent post-polymerization step, were found to decrease (103 °C for the 10 g and 98 °C for the 20 g) compared to B-90 (106 °C). Slower crystallization kinetics, lower crystallinity, second-heating melting points and $T_{d,5\%}$ were monitored for the scaled-up prepolymers. On the contrary, the obtained decomposition temperatures were similar, as they are not that sensitive to molecular weight differences (Table S3).

3.3. Post-polymerization of the scaled-up enzymatically synthesized prepolymer

The prepolymer B-90_20g was selected as the starting material for the following post-polymerization tests due to its adequate available amount and good thermal performance and morphology to permit applying a solid-state post-polymerization step. A precrystallization step was introduced before the main SSP process to eliminate possible sintering issues [55], as it is already used for PET and PLA SSP processes. In particular, the precrystalllization step was conducted at 90 °C under 20 mbar for 2 h (Table 2). The obtained first heating curve was found sharper compared to the starting material (Fig. 8), while an increase of 4 °C in the $T_{m1,}$ reaching 102 °C (Table S2), was monitored, indicating a crystal reorganization with no molecular weight increase. Thanks to this T_{m1} increase, the following SSP process was permitted to be conducted at 95 °C (T_{m1} – T_{SSP} 7 °C) under 20 mbar for 8 h without sintering problems. The SSP product, S8_95, presented slightly upgraded thermal properties compared to the starting prepolymer (Table S2, Fig. 8) indicating polymers of similar molecular weights. Indeed, $\overline{M_n}$ increased slightly to 1900 g·mol^{-1} and $\overline{M_w}$ to 3300 g·mol^{-1}, while dispersity remained almost constant (Table S1).

Since post-polymerization kinetics are highly sensitive to the difference between T_m and T_r, the next approach included the T_{m1} – T_{SSP} reduction up to 2 °C. SSP was conducted at 100 °C under reduced pressure of 20 mbar after precrystallization. However, after the first two hours of SSP, the sample went from the solid to the melt state (S2_M2). This solid-melt transition is a phenomenon already observed during the polycondensation of various nylon salts [56]. By-product (water) accumulation within the reacting particles during polycondensation results in the formation of low melting point areas, which eventually overlap, leading to the complete breakdown of the solid structure. In our case, the reaction by-product (ethanol) and any moisture that the prepolymer surface may have absorbed contribute to the transition from the solid to the melt state. The co-existence of ethanol and water may also synergistically affect the prepolymers' properties; organic solvents like ethanol have been found to swell the polymer matrix and increase chain mobility. Free volume is thus created due to swelling, allowing the water molecules diffusion in the polymeric matrix and thus resulting in hydrolysis [57]. Despite the melt state of the prepolymer, post-polymerization was not significantly favored at 100 °C, not even at longer reaction times (e.g., 4 and 6 h, samples S2_M4 and S2_M6). The highest melting point increase by 12 °C and 4 °C for the first and the second heating respectively was monitored after 4 h of melt polymerization, while the $\overline{M_n}$ and $\overline{M_w}$ increased up to 1900 g·mol^{-1} and 3700 g·mol^{-1} respectively (Table S1).

Thus, in the next step, melt post-polymerization was conducted at 110 °C, under reduced pressure of 20 mbar for 4 h. Before melt post-polymerization, drying at 90 °C, 20 mbar, for 2 h was conducted to remove any possible absorbed moisture from the prepolymer. A sharper curve was obtained in the first heating (sample M4_110, Fig. 8a), and the melting point reached 104 °C, which was maintained during the second heating (Fig. 8b), indicating increased molecular weight values compared to previous samples where T_{m2} was at 102 °C. Actually, $\overline{M_n}$ reached the value of 3000 g·mol^{-1}and $\overline{M_w}$ increased to 6000 g·mol^{-1} with dispersity of 2 (Table S1). The end PBS (enzymatically synthesized and upgraded *via* post-polymerization) was a semicrystalline polymer with a crystallinity degree 46% (Table S2), thermally stable with a

Table 3
Characterization of unloaded (Blank) and naringin-loaded PBS nanoparticles: particle size, polydispersity index (PDI), ζ-potential, encapsulation efficiency (EE), process yield.

	Size (nm)	PDI	ζ-Potential (mV)	EE (%)	Process yield (%)
Unloaded PBS nanoparticles (Blank)	473 ± 53.7	0.76 ± 0.04	−8.5 ± 0.4	–	68
Naringin-loaded PBS nanoparticles	450.8 ± 13.6	0.40 ± 0.03	−12.1 ± 0.2	68	71

decomposition temperature of 393 °C. Even though post-polymerization was conducted in the melt state, the final product was not thermally downgraded thanks to the relatively low temperature and short reaction times in the melt state.

3.4. Encapsulation of naringin in enzymatically synthesized PBS nanoparticles

Using a pure, enzymatically synthesized and upgraded *via* post-polymerization PBS as a carrier in encapsulation systems is challenging and has never been tested to the best of our knowledge. Through the following preliminary tests, the formation of PBS nanoparticles, the encapsulation of naringin and its controlled release were assessed.

Unloaded PBS nanoparticles were first formed to evaluate the effectiveness of the herein applied procedure. The nanoparticles' mean hydrodynamic diameter was about 470 nm (Table 3). The herein formed nanoparticles' polydispersity index was 0.76, indicating a broad size distribution [58], probably attributed to agglomeration. Zeta potential was found to be −8.5 mV. The negative charge of PBS is attributed to its ester end groups C_2H_5O— (pKa = 3.6) that deionize in water, leading to negative charges in the polymer [59]. The absolute zeta potential value (>30 mV) indicates a suspension with aggregation tendency [33,60], in agreement with the aforementioned results.

Naringin-loaded PBS nanoparticles were formed in the next step. The process yield was 71%, similar to the unloaded nanoparticles' (Table 3). The encapsulation efficiency was 68%, slightly higher than analogous encapsulation systems reported in the literature (e.g., naringin encapsulation in PMMA nanoparticles, EE 60%) [61]. The nanoparticles' size and ζ-potential were similar to those of the unloaded nanoparticles, while their polydispersity index (0.4) indicated a more homogenous nanoparticle population.

The following DSC and TGA results verified that the encapsulation was successful. Naringin presented two endotherms, a broader one at 60–130 °C and a sharper one at 160 °C (Fig. 9a). The first one probably corresponds to the loss of water molecules [62], while the second is the melting endotherm. These endotherms are present in the physical mixture of PBS-naringin, weaker and shifted at 120–140 °C and 150 °C but with the same morphology (Fig. 9b). The naringin peaks' shift to higher temperatures in the physical mixture can be attributed to the substances' intermolecular interactions, i.e., hydrogen bonding between

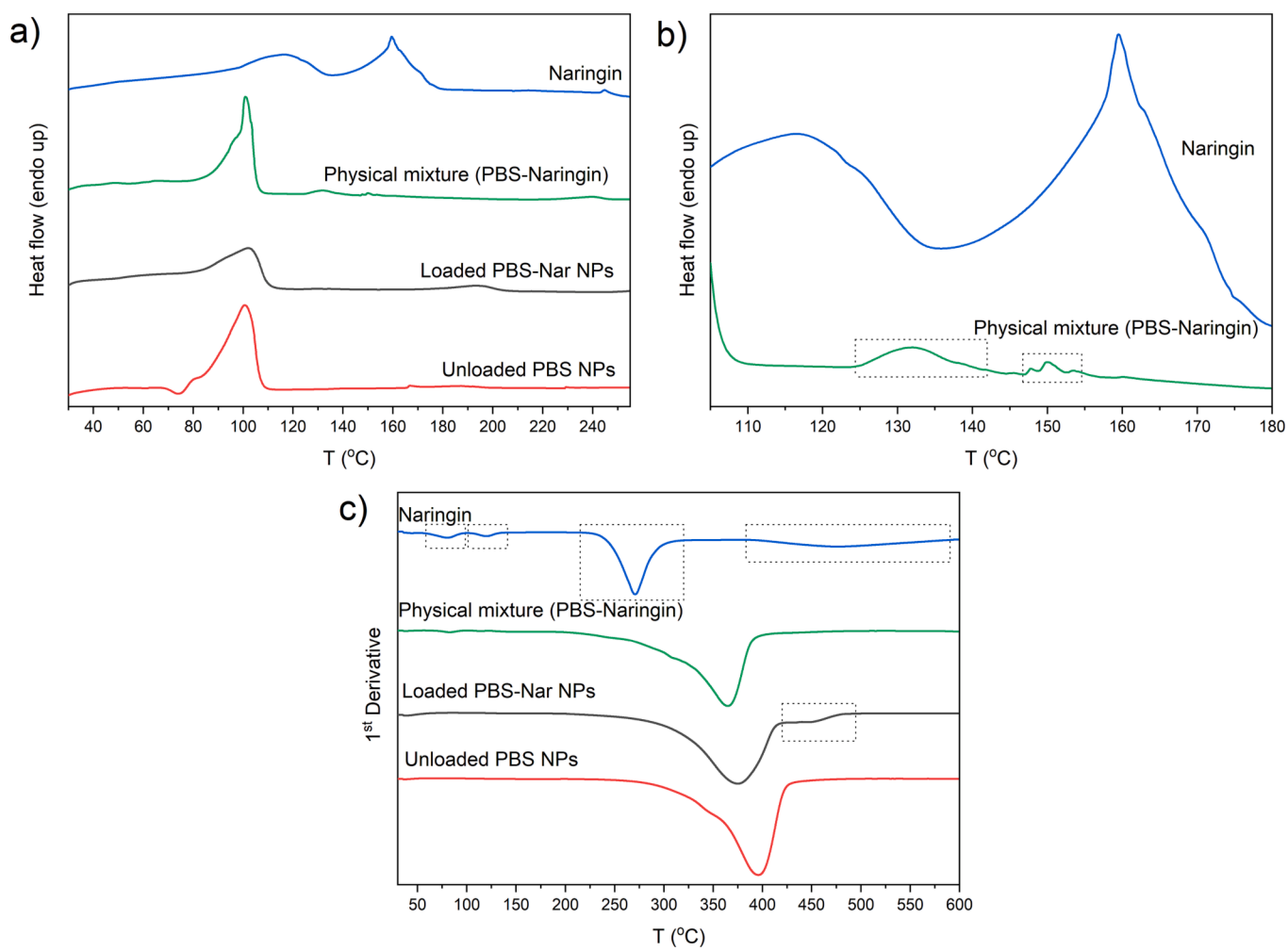

Fig. 9. (a) DSC (1st heating), (b) zoomed DSC (1st heating) and (c) TGA (1st derivative) of Naringin (blue), physical mixture of PBS-Naringin (green), naringin-loaded and unloaded PBS nanoparticles (grey and red respectively).

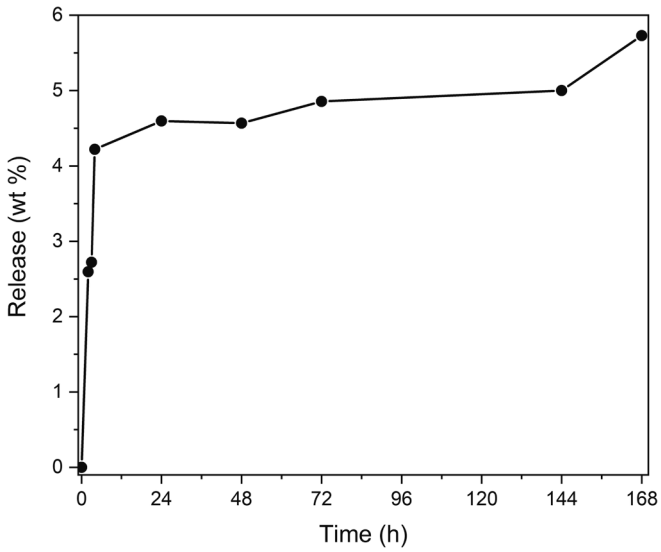

Fig. 10. *In vitro* release profile of PBS nanoparticles loaded with naringin (pH 7.4, T = 25 °C).

the naringin hydroxyl groups (—OH) and PBS ester bonds, requiring additional energy to melt. Based on the loaded nanoparticles DSC curve, naringin was encapsulated as amorphous in PBS nanoparticles. The transition from a crystalline to an amorphous state of an active agent after encapsulation is typical and observed in different encapsulation systems, including naringin-loaded alginate microspheres [63] and curcumin-loaded lipid nanoparticles [64]. This phenomenon could be attributed to the stresses (e.g., mechanical pressure) exerted on the active agent during encapsulation. For instance, the ultrasound energy applied during sonication could cause the naringin crystalline structure deformation. Additionally, the main endotherm of the PBS matrix of the loaded nanoparticles (102 °C) was broader vs. unloaded particles, probably due to the loss of encapsulated naringin's water molecules starting at lower temperatures. The endothermic peak of the loaded nanoparticles' thermogram at 194 °C is attributed to residual PVA [65]. The blank nanoparticles' DSC curve was sharp, and the calculated crystallinity was 54%, while an exothermic peak appeared at 74 °C, revealing an increased tendency of PBS to crystallize. The crystallinity of the matrix is considered a key factor for controlled release, with amorphous polymers facilitating the active agent's diffusion, thus accelerating the release rate [66]. As a highly crystalline polymer (x_c ca. 60%), PBS presents generally low degradation and release rates [67,68].

Regarding TGA results, naringin presented four different decomposition stages (Fig. 9c); the first two (80 and 120 °C) are attributed to the water loss in agreement with the relevant DSC results. The main naringin decomposition occurred at 271 °C, and the final decomposition stage was depicted as a broad peak from 410 to 500 °C. The naringin water loss is also monitored in the physical mixture. However, the main naringin decomposition stages (271 °C and 410–500 °C) are not distinct, confirming PBS and naringin molecules interaction in the physical mixture in agreement with the abovementioned DSC results. The main decomposition peak of the physical mixture is monitored at 365 °C, between naringin and pure PBS decomposition peaks (Table S3), as expected. The loaded nanoparticles presented two distinct decomposition stages; the first is the main decomposition peak monitored at 375 °C (10 °C higher than the physical mixture), indicating a different topology between the loaded nanoparticles and the physical mixture. The second one, monitored at 420–480 °C, is attributed to residual PVA, also evidenced by DSC.

$T_{d,5\%}$ values also confirmed the different topologies of the loaded nanoparticles and the substances' physical mixture. Loaded nanoparticles presented $T_{d,5\%}$ at 269 °C, namely 74 °C higher than the physical mixture ($T_{d,5\%}$ 195 °C) and 35 °C lower than the unloaded nanoparticles ($T_{d,5\%}$ 303 °C). The higher $T_{d,5\%}$ of the loaded nanoparticles compared to the physical mixture, indicated the naringin's uniform distribution within the PBS matrix preventing localized heating, and thus delaying thermal degradation compared to the physical mixture of the substances. The lower $T_{d,5\%}$ of the loaded nanoparticles compared to the unloaded, is due to the water loss of the naringin in the loaded nanoparticles during heating. Overall, the $T_{d,5\%}$ values of all the nanoparticles were decreased compared to the pure PBS matrix ($T_{d,5\%}$ 317 °C, Table S3), even for the unloaded particles, where no different chemical composition exists. This phenomenon is attributed to the size of the particles, as polymers' properties are strongly affected by surface effects and/or chain confinement and mobility and are considerably different for bulk and nano-sized grades [69,70].

The naringin-loaded nanoparticles were then submitted to preliminary *in vitro* release experiments at 25 °C and pH 7.4. These conditions were selected, as food storage (shells, warehouses) or transportation may occur at room temperature, and bacterial growth is enhanced near neutral pH. The release profile is presented in Fig. 10. During the first 4 h, a burst effect was observed during which almost 4.5% of naringin was released. The burst release is usually caused by the active agents' fast desorption at the surface, and a sustained release is expected to follow [67]. Actually, a plateau was reached until 114 h, and an increasing tendency was monitored at 168 h, reaching a value of almost 6%.

The predominant release mechanism affects the release rate, which has been slow in our system (6% for seven days) and should be further investigated as osmosis, diffusion, polymer swelling, and polymer erosion/degradation may contribute to the release process [71]. PBS, though, degrades slowly compared to other biodegradable polymers, and thus, the release is often dominated by the active agent's diffusion from the polymer matrix, making it thus appropriate for very long-term release systems [68]. In this context, Brunner et al. [68] used a commercial PBS grade to form microcapsules and investigated the controlled release of a hydrophilic and a hydrophobic bioactive agent (i.e., bovine serum albumin (BSA) and all-trans retinoic acid (atRA), respectively). BSA was released at 6% after 3 weeks and atRA at 9% after 42 days. Based on the herein results, it could be considered that diffusion is the predominant mechanism until 144 h, and PBS hydrolysis is the following predominant mechanism, becoming more intense after 144 h. Overall, the encapsulated naringin release confirms that the enzymatically synthesized PBS can be used as a matrix, especially for long-term release systems, despite its low molecular weight.

4. Conclusions

The present study aspires to provide a sustainable method to produce a high-purity PBS, able to be used in the biomedical and food packaging sector. PBS prepolymer was synthesized *via* enzymatic polymerization using immobilized *Candida antarctica* Lipase B as a biocatalyst in a solvent-free system. The enzymatic synthesis was conducted in two stages, with the first at milder conditions (40 °C, atmospheric pressure, 24 h) to minimize possible monomers' losses. The second stage's conditions (reaction temperature, pressure, time) were thoroughly investigated, and 90 °C, 20 mbar, 2 h were indicated as the optimum, providing PBS of $\overline{M_n}$ 2500 g·mol^{-1} and T_m 105 °C. Thanks to the process's simplicity, it was scaled up, and 20 g of enzymatically synthesized PBS were received for the first time to the best of our knowledge. The scaled-up prepolymer's properties were slightly lower due to mass transfer limitations that are more intense on large scales. On that basis, the scaled-up prepolymer was submitted to different post-polymerization methods, including SSP and melt post-polymerization, to increase the molecular weight. Melt post-polymerization was found effective and resulted in a 76% increase in the $\overline{M_n}$. The upgraded prepolymer was a high-purity PBS grade appropriate for biomedical and packaging

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applications where controlled MW and crystallinity strongly affecting the degradation rates are needed. In that context, an antioxidant, naringin, was successfully encapsulated, indicating the product's suitability to act as a matrix in encapsulation systems. The preliminary herein conducted *in vitro* release experiments indicated that after 168 h, almost 6% of naringin was released and confirmed that the herein enzymatically synthesized PBS is a promising candidate carrier in controlled release systems.

Author statement

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

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